How to Increase the Success of Innovative Start-Ups with Intellectual Property: An Introductory Overview

Six Views but One IP to Rule Them All

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Do not be driven into darkness.

1 Start-Ups and Their Need for Capital

Have you ever been in a situation where you have had a brilliant idea and later hear about a start-up that is having incredible success with a similar idea? This happens to many people who have great ideas, but they don't immediately start their own business. There are numerous books about founders, start-up strategies, framework conditions, and the start-up ecosystem. Many ingredients are important for the successful establishment of a start-up. One important ingredient that contributes to the success of start-ups is intellectual property, which must be managed regardless of whether it is protected or not.

The last 10 years have been characterized by a rising interest in start-up ventures and high investments in start-ups worldwide. New types of investors have entered the market, such as hedge funds, corporations, or the "crowd" besides the classical investors such as venture capitalists and private equity firms. In 2021, start-ups

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raised US$628 billion which more than doubled compared to the previous year, and more than 528 private firms became "unicorns"[1] (CB Insights, 2022b).[2] This was favored by expansionary monetary policy and a high volume of money in circulation in the largest markets. Regardless of market conditions, there are snowstorms – ups and downs which again impact the enthusiasm of investors.

Current trends and developments affecting the start-up market include:

- After the fat years, dealmaking is slowing down;
- More new businesses are springing to life;
- Multiple start-up hubs are rapidly evolving around the world; and
- China's ambitious plan to become the world's next center of innovation.

Fat Years of Dealmaking Are Slowing Down Recent trends indicate a turning point. Crises, wars, supply-chain shortages, sharply rising inflation rates, which is counteracted by rising interest rates, erode the value of future cashflows and lead to a reassessment of investments in the start-up market. The quarterly CB Insights report found that global venture funding dropped in Q1 2022 to US$143.9 billion raised across 8,835 deals. This is a 19% decrease from the previous quarter's funding, as shown in Fig. 1.[3] Global venture funding continued to decline sharply in Q2 2022 to US$112.6 billion and in Q3 2022 to US$74.5 billion (CB Insights, 2022b). Also, global medium valuation of late-stage deals dropped to US$1.054 billion. Despite this revaluation, there are numerous start-ups that remain highly valued. In Q1 2022, the three highest valued unicorns worldwide were: *ByteDance*, founded by *Zhang Yiming*, the Chinese tech company which owns *TikTok*, valued US$350 billion; *Ant Group* (China), founded by *Jack Ma*, valued at US$150 billion; and *SpaceX* (United States), founded by *Elon Musk*, valued at US$100 billion (CB Insights, 2022a). Solid start-up valuations based on promising business models, technologies, and intellectual property will persist, and start-ups with well-strategized intellectual property (IP) concepts will find it easier to raise capital (Süzeroglu-Melchiors & Hammans, 2022).

More and More Start-Ups Are Springing to Life In 2021, the number of newly formed companies was 15% higher than the average before the pandemic (The Economist, 2022). According to the global start-up ecosystem's 2022 rankings, most start-ups are established in the *software* and *data* industry, with 31.8%, followed by *healthtech* with 12.7%, *social* and *leisure* with 10.4%, and *fintech* with 10.2% (Crunchbase, 2022).[4]

[1] Start-ups with a valuation greater than $1 billion.

[2] https://www.cbinsights.com/research/report/venture-trends-2021 (accessed October 24, 2022).

[3] https://www.cbinsights.com/research/report/venture-trends-q1-2022 (accessed October 24, 2022).

[4] Based on a sample of more than 100,000 start-up entities. https://about.crunchbase.com/blog/trends-global-startup-ecosystem-2022 (accessed July 07, 2022).

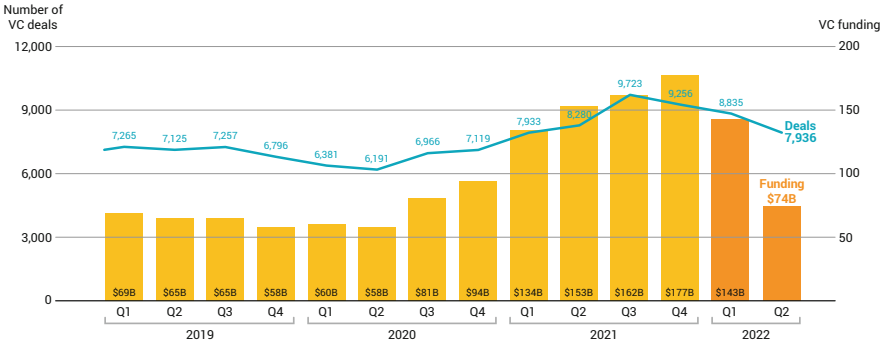


Fig. 1 Worldwide venture capital funding 2019–2022 (source: CB Insights)

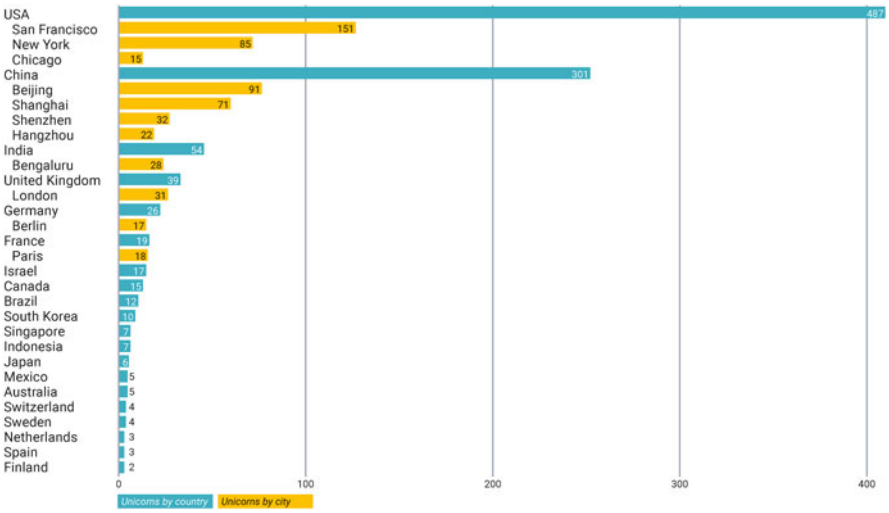


Fig. 2 Top 20 ranking of countries and cities with the most unicorns in 2021 (source: Statista. https://www.statista.com/statistics/1062457/cities-highest-number-of-unicorns, accessed October 24, 2022)

Multiple New Start-Up Hubs Are Evolving Around the World In recent decades, *Silicon Valley* has been the engine of start-up activities and the birthplace of many tech start-ups. A number of large cap companies[5] were founded in Silicon Valley, such as *Apple*, *Hewlett-Packard*, *Intel*, *Google*, and *Uber*, while others, including *Facebook*, have moved to the Valley shortly after being founded. Today, Silicon Valley is still the most fertile soil for start-ups, but other start-up hubs around the world are evolving and catching up, as shown in Fig. 2.

Reasons for this development are the diffusion of investments and funds into big cities and government support programs which foster growth in start-up hubs.

[5]Large cap refers to companies with a market capitalization of more than $10 billion.

Besides *Silicon Valley*, the largest start-up hubs today include *Tel Aviv*, *Beijing*, *London*, *Berlin*, *Singapore*, *Bengaluru,* and *São Paulo*.

In 2021, there were a total of 1,058 unicorns in 42 countries. Some 487 unicorns were headquartered in the *United States*, including 151 in *Silicon Valley*. Other countries in the top five rankings were *China* with 301 unicorns, *India* with 54 unicorns, the *United Kingdom* with 39 unicorns, and *Germany* with 26 unicorns, as shown in Fig. 2 (Statista, 2021a, 2021b).

China's Ambitious Plan to Become the World's Next Center of Innovation The 14th Five-Year Plan (FYP) (2021–2025) announced in March 2021 *China's* ambition to become a world-leading innovator by 2035. The 14th FYP proposes to foster high-quality development by supporting innovation and upgrading human capital. Research and development spending is planned to increase by at least 7% every year between 2021 and 2025 (East Asia Forum, 2021).[6] These ambitious plans are also reflected in the development of the start-up scene. China's government offers extensive subsidies to start-ups, and state-supported incubators are driving the domestic e-commerce start-up scene, increasingly turning China's digital founders into genuine Silicon Valley competitors.

2 Why IP Actually Does Matter: The Status Quo

It may be common sense in practice and academia that intellectual property is broadly used by established companies for creating an innovative edge and sustaining their competitive advantage (Gassmann & Bader, 2017; Gassmann et al., 2021; Grandstrand, 2018; Holgersson et al., 2018; Tietze, 2019). And surely many start-ups have heard the advice "file your patents." But in reality, start-ups are often overwhelmed with intellectual property matters. Reasons are diverse, such as the lack of IP knowledge, no dedicated budget, limited capacities, no access to IP software tools, etc. From an IP perspective, start-ups are at a disadvantage per se. This can be compared to a 100-meter race, where the IP-experienced companies already have a 10-meter lead, wear better running shoes, and thus have a clear advantage. Despite the general advice to file valuable patents, start-ups often prefer a risky but fast approach based on confidentiality. Therefore, this compilation is designed to draw attention to IP issues and to show why IP actually does matter, for reasons such as:

- Few start-ups protect their IP, but these are more successful;
- Patents are signals to stakeholders that the start-up is worth investing in;
- Patents are a weight on the negotiating table;
- Functions of patenting support the start-up.

[6] https://www.eastasiaforum.org/2021/05/08/chinas-plan-to-become-a-world-leading-technology-force (accessed October 24, 2022).

Few Start-Ups Protect Their IP, but These Are More Successful Only a few start-ups protect their IP rights, but these are significantly more successful than start-ups that do not protect IP rights. According to a recent study by the EPO and EUIPO (2021), fewer than 9% of European SMEs rely on protected IP rights, but this subset of companies appears to generate 68% higher revenues per employee than SMEs without IPR portfolios. There is clearly a positive relationship between patenting start-ups' innovations and economic performance.

Patents Are Signals to Stakeholders that the Start-Up Is Worth Investing in It is widely known that patents are a proxy for innovation. Patents can be seen as proof that the technological idea is new and innovative and that a start-up is investing precious funds in patent applications. Patents signal the legitimacy to move from conception to commercialization. Most investors perceive intellectual property rights, such as patents, patent applications, utility models, and trademarks, as an indicator for quality and value. A study utilizing data from 468 early-stage Canadian start-ups shows that patents clearly increase the likelihood and amount of receiving venture capital because they signal the quality of the venture (Vo, 2019). This perceived signal has its validity, as there is clearly a positive relationship between patenting start-ups' innovations and economic performance (EPO and EUIPO, 2021). Furthermore, patents have a positive effect on other financing instruments such as governmental funding programs and debt financing. Recent studies show that patents increase the likelihood of receiving governmental grants (Islam et al., 2018), and patents are assets that can be used as collateral for loans. For example, in 2016, there were 117,262 patents and patent applications serving as collateral in secured financing by banks in the United States (Nguyen & Hille, 2018). Another U.S. study shows that 36% of technology start-ups in the computer software, semiconductor, and medical devices sector used patent-backed financing to receive loans (Hochberg et al., 2018).

Interestingly, signaling effects of patents work like a chain effect. Patents make an essential contribution to the success of early-stage funding, and prior funding positively affects follow-up financing because prior funding certifies firm quality to investors and reduces information asymmetries. A study based on 221 business plans and project descriptions of start-ups that ran equity crowdfunding campaigns on the British platform Crowdcube in 2017 and 2018 found that prior financing positively affects crowdfunding campaign success (Kleinert et al., 2020). Also, start-ups which received a prior grant were 12% more likely to acquire subsequent venture capital (VC) funding (Islam et al., 2018). This clearly shows that patents, have a positive effect on the raising of capital throughout the venture life cycle. As an asset, they are used as collateral and an indicator for the venture quality and help start-ups to acquire funding and overcome the well-documented valley of death.

Patents Are a Weight on the Negotiating Table Start-ups are often faced with the situation that they are dealing with experienced investors and large corporations that wield effective negotiation tactics. Patents are a unique selling point here because they prohibit others from using the protected technology. Protected intellectual

property rights strengthen the negotiating position vis-à-vis investors and large corporations and can make a significant contribution to successful negotiations, securing orders, and achieving a return on the invested (ROI) research and development.

Functions of Patenting Support the Start-Up Often early-stage ventures decide against patenting in favor of non-disclosure agreements (NDAs). This can make sense if the patenting process would take too long compared to the short technology life cycle or if the probability of obtaining a patent is low. On the other hand, patents are a good alternative to NDAs because they provide stronger control over information disclosure. In any case, start-ups should weigh the options carefully. The functions of intellectual property are not only to protect one's own innovations from imitation, but also to ensure commercial exploitation through licensing, to create collaborations through cross-licensing, and to guarantee freedom to operate (WIPO, 2021). A series of EPO case studies in 2017[7] shows impressively that start-ups can leverage their IP rights to grow and expand by licensing in and out, establish partnerships, and prevent others from imitation, therefore securing high profit margins (see also EPO and EUIPO, 2019; EPO, 2022).

There are diverse ways to leverage IP and take advantage of positive effects such as signaling effects. It takes more to harness the full potential of IP than just to manage a brilliant idea. Start-ups are strongly advised to treat IP as a key ingredient, creating and protecting value for their company. We take a detailed look at this ingredient – intellectual property – from diverse stakeholder views, state of the art in research and in practice, and we address many important questions within the start-up ecosystem.

3 Six Main Views That Allow Differentiated Insights on Intellectual Property Management for Start-Ups

This book was written to fill what we believe to be a gap in the field of intellectual property management for start-ups. The fact that IP is important for young ventures as well as established companies is undisputed. While established companies spend both financial and professional resources to give this topic the attention it deserves, start-ups struggle to manage their intellectual property. There are various reasons for this, such as:

- Lack of awareness of valuable intellectual property;
- No knowledge about the various IP rights and conditions and ways to become the proprietary owner of intellectual property;
- Obliviousness to develop intellectual property strategies;

[7] https://www.epo.org/case-studies (accessed on July 04, 2022).

- Limited experience in how to exploit intellectual property and defend themselves against third parties;
- Ignorance of the fact that legal consequences can be an existential threat for start-ups; and
- Finally, start-ups are frequently overwhelmed with the complexity of the intellectual property domain.

This list could go on for a long time. We address precisely these problems, ask the main questions about IP, and provide answers in a best-practice, user-friendly way. Literature on start-ups does not only speak about individual start-ups but often refers to the critical infrastructure, namely the start-up ecosystem, which is particularly important for young companies. We make use of stakeholder perspectives that often have a major influence on the success and failure of start-ups. There is a long way to go from invention to commercialization of the innovation, and in between, there are many steps and activities which involve different stakeholders. The most common business questions and concerns that arise include:

- What type of IP is needed to create and capture value for sustainable success?
- What should be proprietary IP and what can be shared IP?
- Which signals does the start-up send to the market with IP?
- What can be publicly disclosed and what should be kept confidential?
- Which inventions are patentable and who can help?
- How can inventions be best protected?
- What are the benefits of registering IP?
- What budget is needed for IP protection, for how long, and how can it become part of the investment plan?
- What is to be done for a reasonable risk mitigation?
- Does the start-up infringe others' IP?
- Can IP be used to gain and grow assets for start-up valuation and exit?
- How does academic technology commercialization work?
- What are current developments and evolving institutional programs that support start-up IP activities?
- Which databases and methods can be utilized for strategic and effective IP management?
- Which best practices for managing intellectual property can support start-ups' IP activities?

In this compilation, we address all these questions contributed by some of the world's most renowned experts in IP and related fields.

We identified six main categories of stakeholders that define the space for IP management for start-ups through active, passive, or advisory intervention. These six stakeholders enrich the volume with their own perspective, experience, interest, and stories.

Therefore, the book is structured into six parts each reflecting a range of perspectives from a specific stakeholder group in the start-up ecosystem. These six main views cover a wide range of all relevant topics on IP management for start-ups,



Fig. 3 Six stakeholder views within the start-up ecosystem on IP (authors' own figure)

as shown in Fig. 3. This structure reflects: the start-up entrepreneurs, encounters with investors, corporations, university technology transfer offices, global IP offices, and advisors, i.e., consultancy and specialized legal practice services. The six parts of the volume are prepared by selected expert contributions based on their track record and profound expertise in IP and related fields.

The six viewpoints that form the connecting thread of this book provide a broad picture of diverse opinions, which are partly overlapping but also partly conflicting. There is no "one and only" management theory that perfectly reflects all aspects of IP management for start-ups. Thus, this compilation is to be understood as a bridge between academic aspects, existing literature, and applied practices. The six views contain IP topics from all over the world and cover a wide range of different technology sectors and industries. It is this diversity that provides the added value for the reader.

To conclude, some start-ups and their stakeholders might still be floating under an umbrella of foggy ignorance; others might be paralyzed into inaction. The acronyms *FOMO* and *SOBS* come to mind in this situation: You don't want to spend money and time on IP just for *fear of missing out* on a potential opportunity, but you don't want the *shame of being suckered* either if you find your company value eroded for not having duly invested in IP. The decision shouldn't be speculative but rather deliberate, and we hope this book will help start-ups and their ecosystems tease out what is to be done.

In the following, we introduce each partition with an explanatory summary that is provided based on the chapter abstracts. We conclude with a condensed overview of the covered topics, industries, technologies, regions, and countries by the six stakeholder views provided in Table 1:

Part I: The Start-Up's View

The Start-Up's View addresses the most common questions and concerns that founders have regarding protecting IP, avoiding infringement of others' IP, exploiting IP, as well as gaining and growing assets for valuation and exit. Authors share learnings coming from Israeli start-ups and their specific ecosystem, also known for the high relevance of its cybersecurity sector; from the German climate tech and sustainability sector where IP proves to be a new value driver; from the Australian global mining industry where innovation relies on the mining equipment technology services sector; and from the concentrated Brazilian information technology and software industry, which maintains strong links to the financial capital and "matchmaking" region Silicon Valley.

Israeli Start-Ups and Ecosystem has been contributed by Roy Melzer, a patent attorney and the Head of the Computing Technologies Department at Ehrlich Group, Tel Aviv, Israel. He summarizes: Israel is referred to as the "Start-up Nation" due to the population's unique entrepreneurial DNA, the massive research and development activity, the high level of venture capital activity, and its substantial state funding that nourishes highly successful start-ups across industries. Most start-ups in Israel and worldwide have a defined life cycle of different phases. Each phase has unique characteristics that directly affect the strategy for intellectual property in general and patents in particular. His chapter describes the adaptable IP strategy, different intellectual property matters that are typical for each start-up stage, and demonstrates best practices for handling these matters (Melzer, 2023).

IP as a New Value Driver for Sustainable Start-Ups has been contributed by our authors Dr. Sven Gensler, Chief Financial Officer, Dr. André Brosowski, Chief Sustainability Officer, and Philipp E. Hammans, CEO, all three at JenaBatteries, Jena, Germany. They summarize: Total investment for climate tech firms, the pace of start-up development, and the average funding size have all increased during the last 10 years. Early-stage venture financing for climate tech start-ups totaled US$418 million in 2013. However, overall venture capital surged by more than 3.750% in 2019 to $16.1 billion. While it is fashionable to be sustainable, little is known about how a business model benefits and successfully implements sustainable technology. Theoretical and realistic sustainability approaches share several similar characteristics: Enhancing sustainability often requires transition, innovation, or adaptation. The question is: What is the difference between "sustainable technology" and "sustainable innovation"? The ability to innovate in sustainability is a requisite business capacity, whether linked to minor incremental measures or extreme, revolutionary developments. There is, however, a lack of transparency and consistency in the terminology and a lack of demonstrated scientific grounding in value creation. To valuate a new venture or start-up is a critical and long-debated question in entrepreneurial financing. Determining a start-up or any other company's economic valuation is one of the more challenging and vital discussions a founder can have with institutional investors. One of the greatest assets of a start-up and the cornerstones in innovation is intellectual property and intellectual property rights. Their chapter discusses how sustainability, defined through the Sustainable Development Goals (SDGs) metrics, is used as a new driver in the valuation of the technology and intellectual properties in order to reflect the true value of a

company's intellectual property when looking for financing through, for example, collateralization of its IP (Gensler et al., 2023).

Mining IP: A Story of Start-Ups in the Global Mining Industry has been contributed by our authors Dr. Dallas L. Wilkinson, Regional Managing Director Asia Pacific, and John Walker, Senior IP Consultant and patent and trademarks attorney, both from Dennemeyer Group. They summarize: Two SME companies (considered start-ups) which operate in the Mining Equipment Technology Services (METS) sector of the Global Mining industry have been studied to understand the role that their IP, along with their associated IP strategy, has played in their success. With both companies starting to expand internationally, their IP strategies have become increasingly important. Both companies are reliant on technology and thrive due to a focus on and understanding of IP. Generation and protection of IP has allowed these organizations to continually grow by maintaining a strong awareness of IP across their organizations. Key insights from the case studies include the importance of IP awareness, training, non-disclosure agreements, the critical nature of a company strategy composed of robust technology and IP strategies, the importance of utilizing all relevant types of IP protection, complexities added through collaboration, taking a proactive approach, and finally, the absolute requirement to remain focused on the commercialization goals while being very conscious of IP. The clear message is that IP is a valuable asset which needs nurturing and must be actively managed throughout the entire technology life cycle and interconnected commercialization phase (Wilkinson & Walker, 2023).

IP and Start-Ups in the Context of Florianópolis, the Silicon Island of Brazil, has been contributed by our authors Dr. Thiago Marques, Professor at Senac-SC and Chief Marketing Officer at Zucchetti, and by Dr. Andressa Pacheco, Professor at Universidade Federal de Santa Catarina, both from Florianópolis, Brazil. They summarize: The management of a company's intellectual property is a key issue in its strategic planning and long-term competitiveness. This issue becomes even more relevant when it comes to start-ups given the inherent innovative nature of this type of business. Despite its clear importance, developing countries such as Brazil suffer from a lack of structure, delays, high costs, and legal insecurity in the matter of patents and intellectual property, which makes it difficult for start-ups to obtain the necessary registrations. One of the most common ways to solve the problem in this case is to look for ways to facilitate the process available in the local innovation ecosystem. Florianópolis is one of the most important start-up centers in Brazil and serves as a benchmark for cities and governments in developing countries that want to create strong innovation ecosystems (Marques & Pacheco, 2023).

Selected Takeaways (the Start-Up's View)

- The clear message is that IP is a valuable asset which needs nurturing and must be actively managed throughout the entire technology life cycle and interconnected commercialization phase.

(continued)

- A typical start-up will go through certain stages in its life cycle; each has its own characteristics and challenges that directly affect the strategy for intellectual property in general and patents in particular.
- The "Israeli way" of creating extremely successful start-ups involves a great deal of creativity, thinking outside the box, perseverance, and the ability to fail, learn, and repeat.
- The connection with the local ecosystem facilitates the understanding of the importance of legal protection of the innovation and can bring immediate benefits to the start-up seeking this support.
- A pre-deal IP valuation project is an optimal tool for risk assessment as the vital part of an IP valuation project consists of identifying legal, economic, and technical risk potentials within a company's intangible asset portfolio.

Part II: The Investor's View

The Investor's View addresses IP as a proxy for innovation and future growth as well as how to tackle IP during the due diligence and valuation processes. Authors examine the significance of IP in early-stage start-up investment decisions on the basis of insights from the United States, Israel, and Switzerland, and the role of IP and intangible assets in tech private placements from a Silicon Valley investor's perspective, as well as the relevance of IP for the investment process in start-ups and early-stage companies from a Canadian angle. Non-equity-based capital raising opportunities for start-ups on the basis of leveraging their intellectual property and intangible asset portfolios are revealed by front-runner BDC Capital.

IP Is Paramount: The Significance of IP in Early-Stage Start-Up Investment Decisions has been contributed by Dr. Ulf G. Schaberg, an IP consultant at Intellectual Property Business Partners, and IP Manager of the AO Foundation, a not-for-profit foundation in Davos, Switzerland, who has interviewed three different investors from the United States, Israel, and Switzerland. He summarizes: Venture capital investors play an important role in the start-up ecosystem, particularly during early stages. To understand how investors think about intellectual property and what role it plays in their investment decisions, this chapter provides interviews with three experienced VC experts. The interviews confirmed, unsurprisingly, that intellectual property is a very important aspect for VC investors in their investment decision-making process. Investors have clear expectations in certain areas while they are less concerned in others, and clear preferences exist as to how and when intellectual property should be in the focus during a typical assessment process. A stepwise approach is recommended for founders to gradually and efficiently climb up the ladder of IP savviness, to be prepared to meet investor expectations in the next pitch to unlock access to capital (Schaberg, 2023).

The Role of IP and Intangible Assets in Tech Private Placements has been contributed by our authors Praveen Shah, Founder and CEO, and Pallavi Shah, Managing Director, Transactions, both of Mobity LLC, a Silicon Valley boutique

firm that, founded in 2002, is a strategic advisory, IP transaction broker, and venture incubator/accelerator headquartered in Mountain View (Silicon Valley), California, United States. They summarize: We examine the role intellectual property and intangible assets play in technology private placements from the viewpoint of an investor. We also incorporate the perspectives of various other stakeholders involved in launching, financing, operating, scaling, and exiting technology ventures. To provide actionable insights our readers can use on their next deal, whether as an investor or an investee, we attempt to engage and educate our readers through illustrative case studies based on our combined six decades of relevant and personal experiences. We also draw upon the collective wisdom of various industry practitioners we have worked with during our careers (Shah & Shah, 2023).

What's the Big Idea? The Crossroads Between Investment and Intellectual Property has been contributed by our authors David Durand and Cédrick Mulcair, both Avocats/Lawyers at David Durand Attorneys Corporation Inc., Montréal, Canada. They summarize: The purpose of our chapter is to assist readers in demystifying the investment process in start-ups and early-stage companies. The authors address this topic from both a start-up and investor perspective and provide further insight on determinant factors of the investment decision-making process. While focusing on the Canadian regulatory landscape, the authors emphasize the importance of establishing an intellectual property strategy (MVIP) early on, enhancing the ability of start-ups to attract capital from investors, as well as to foster competition and productivity (Durand & Mulcair, 2023).

Intellectual Property-Backed Financing has been contributed by Lally Rementilla, Managing Partner at BDC Capital, Toronto, Canada. She summarizes: Start-ups can access non-dilutive financing by leveraging their intellectual property and intangible asset portfolios through intellectual property-backed financing. The author uses her own experience to show how intellectual property-backed financing works, its costs and benefits, and how start-ups can prepare themselves better to access this type of financing (Rementilla, 2023).

Selected Takeaways (the Investor's View)

- If you want VC funding, take IP seriously.
- To get there, work up the ladder of IP savviness.
- Align your IP strategy with your business strategy and model early on and find ways for them to complement one another.
- It is critical to have a strong "product-market-moat" fit, strong IP strategies, and IP management culture that are aligned with business strategy.
- IP can be a good proxy signal in the early (but less so in the later) stages.

Part III: The Corporation's View

The Corporation's View addresses corporations' start-up strategies that manage IP as an important value driver for innovation and growth along with getting access to new technologies and markets. Insights come from the automation and industrialization

industry, including the platform economy, internet of things (IoT) and Industry 4.0, i.e., the global technology powerhouse Siemens and its Siemens Technology Accelerator, the information and communications technologies industry with learnings from Nokia, and the pharmaceutical, life sciences and crop science industry sector, i.e., Bayer. Readers will appreciate insights from different cultures of international and global companies, including learnings from the Japanese start-up ecosystem and corporate culture that reveal how Japanese companies bridge the cultural gap between start-ups and corporations.

IP as an Important Value Driver for Start-Up Strategies in Established Companies has been contributed by our authors Beat Weibel, Chief IP Counsel Siemens, and Dr. Rudolf Freytag, Head of Licensing at Siemens and CEO of Siemens Technology Accelerator, Germany. They summarize: Established companies apply two types of start-up strategies to drive innovation and growth: With outside-in start-up strategies, they get access to external start-up innovations; with inside-out start-up strategies, they build independent new innovative businesses. These two strategies are fundamentally different, but they have in common that IP becomes an important value driver when a value-driven IP strategy is pursued. Based on their own experience, the authors show how this approach works in practice and theory (Weibel & Freytag, 2023).

Strategic Uses of IP: Applying Learnings from Large Companies to Start-Ups has been contributed by Sonja London, General Counsel and Licensing Executive at TactoTek Oy and former Head of Consumer Electronics Licensing at Nokia, Finland. She summarizes: Established technology companies normally have an intellectual property strategy in place. This may include at least elements of patenting, trademark filing, IP risk mitigation, and a plan for using IP in their business. For building an effective IP strategy, these companies have considered many strategic ways to exploit their IP. Start-ups can learn a lot from larger companies' approaches to IP. Drawing from her own experience, the author explains the strategic uses of IP from a large company's vantage point, as well as how to apply this knowledge to the IP strategy of start-ups. The aim is to provide start-ups with views and ideas for developing their own IP strategy. Her chapter is based on extensive experience with companies in mobile devices and related technologies, consumer electronics, and general electronics companies. In this context, IP means specifically patents and does not discuss trademarks or branding. The larger context of standard essential patents is also excluded and only addressed on a high level (London, 2023).

How the Life Sciences Industry Handles IP in Start-Ups has been contributed by Dr. Jörg Thomaier, Global Head of IP Bayer Group, Managing Director Bayer IP, Germany. As Head of IP for a global blue chip, he shares his views on how a start-up should take care of its intellectual property, strategically build up a portfolio, and avoid certain kinds of mistakes. His chapter draws on extensive first-hand experience when investigating start-ups to prepare for investing in, collaborating with, or taking over such entities. He also offers insights and takeaways for investors on what to look for and expect from a start-up and its management when getting ready to fund and invest. Some hints for licensors, specifically universities and other publicly funded

entities, are also included. The chapter is a valuable read for anyone interested in gaining insights and advice from a leading IP professional (Thomaier, 2023).

Mind the Gap: Business and Cultural Aspects for Start-Up Collaboration in Japan has been contributed by Robert Alizon, Chief IP Officer of Fairway Family Office, Zurich, Switzerland. He summarizes: Corporations seek partnerships with start-ups to enhance their innovation efforts and to introduce disruption in their market, provided a strong IP portfolio supports the opportunity developed by the start-up. It is thus important for the start-up to apply IP management principles that will lead to successful partnerships. Namely making sure that the start-up owns all IP related to their proposition, the IP secures a strong exclusivity to keep the competition at a distance from the market it has developed, and avoids infringing the IP of other players in the market will attract corporations. His chapter will review these points with an emphasis on the business culture gap that exists between a start-up and a Japanese corporation that is often seeking partnerships with overseas start-ups to compensate for the modest number of start-ups in the Japanese ecosystem (Alizon, 2023).

Selected Takeaways (the Corporation's View)

- IP has become an important value driver of start-up strategies of established companies especially if value-driven IP strategies are applied.
- Using IP strategically allows companies to build shareholder value by securing optimal freedom of action in the relevant market and enabling an advantageous position over competitors.
- IP strategy implementations for start-ups include considering different ways of value creation and assessing risk and planning for its mitigation.
- Ensure that IP rights are unequivocally owned by the start-up and respect employee invention law.
- Make sure to acknowledge cultural gaps between start-ups and corporations.

Part IV: The University's View

The University's View brings an understanding of how technology transfer and the licensing of intellectual property works from a university's perspective. Authors examine the successfully applied practices and case studies from New York City's U.S.-Israeli Jacobs Technion-Cornell Institute at Cornell Tech, South Africa's Western Cape-based Stellenbosch University illustrating the technology transfer journey from invention to commercialization, and Europe's largest entrepreneurship center at the Technical University of Munich, Germany. In addition, we share the results of a study performed by the renowned Swiss university ETH Zurich on how to manage IP-related tensions between universities and spin-offs to navigate through the agreement finding process.

Experimentation in Academic Technology Commercialization has been contributed by Dr. Fernando Gómez-Baquero, Director of Runway and Spinouts, Jacobs Technion-Cornell Institute at Cornell Tech, Cornell University, New York,

United States. He summarizes: Technology commercialization is now one of the cornerstones of the experience in the academic environment. Universities and colleges are now highly dedicated to understanding and giving priority not only to creating new technologies but also to commercializing them. His chapter describes some of the experiences and empirical evidence that we received as we were building the Runway Startup Postdoc Program at the Jacobs Technion-Cornell Institute at Cornell Tech. It explains three broad steps of technology transfer experimentations and flips the traditional technology and transfer office (TTO) model. Experimenting should be a natural component of incubator programs and the most effective way of producing meaningful change in technology transfer systems. His recommendation to academic incubators and TTOs is to embrace experimentation and build a technology transfer culture that is less concerned about the traditional output metrics (i.e., patents, licenses, revenue) and is more geared toward providing people with the resources, connections, and knowledge they require to create a start-up (Gómez-Baquero, 2023).

A University Perspective with a South African Flavor has been contributed by Dr. Madelein Kleyn, Director Technology Transfer, Stellenbosch University, South Africa. She summarizes: Technology transfer of research emanating from a publicly funded research institution operates within a strict legislative and intellectual property policy framework. Understanding the complexities of the university environment is challenging but rewarding, as Silicon Valley and Stanford University, along with many other university technology and science parks, have proven over decades. Not all university innovation and technologies are suitable for licenses or start-ups. The choice of the model for commercialization is complex and intriguing. Her chapter shares some insights (Kleyn, 2023).

The Technology Transfer Office as Facilitator Between Researchers and Investors: A German Perspective has been contributed by our authors Dr. Alexandros E. Papaderos, Head of Innovation, Technical University of Munich, and Oliver Bücken, Investor and Senior Adjunct Lecturer, UnternehmerTUM, Europe's largest entrepreneurship center, Garching, Germany. They summarize: Higher education institutions face the challenge to fulfill their role in the so-called "third mission," next to research and teaching. One major part of this third mission is technology transfer and, consequently, the creation of technology-oriented and knowledge-based spin-offs. Universities and non-university research institutions can succeed in this particular mission by providing a supporting culture and suitable policies, which allow spin-offs to become important drivers of innovation. The journey from research results to the development of products and services to market maturity needs intensive research and thus time and money. An efficient intellectual property management by the university's technology transfer office plays an indisputably crucial role. Founders must be aware of the rights of third parties. Bringing patents into a company should be the normal case. In reality, the contribution of patents plays a huge role when it comes to the distribution of shares. Attracting venture capital and fresh money for scaling goes along with the necessity of freedom to operate. Having shareholders in the company who hold patents privately prevents growth capital to fuel the company. In their chapter, they present the views of a

university TTO, a university researcher/founder, and an investor/practitioner on the importance of IP for university spin-offs (Papaderos & Bücken, 2023).

Managing IP-Related Tensions Between Universities and Spin-Offs has been contributed by our authors Prof. Dr. Stephan M. Wagner, Chair of Logistics Management, and Alexander A. Fink, both at the Department of Management, Technology, and Economics, ETH Zurich, Switzerland, as well as by Joshua Arbter, McKinsey & Company. They summarize: Intellectual property is a critical asset for the growth of university spin-offs. Since the IP usually belongs to the parent university, spin-offs must find an agreement on how to access it. However, there is little evidence on how universities' technology transfer offices and spin-offs find this agreement. Their study aims to shed light on this process. Drawing on four Swiss and German TTOs and 12 related spin-offs, the authors provide insights into navigating through this process. The data reveal that both parties face four critical tensions while negotiating: (1) TTO's success might hamper spin-offs' success; (2) TTO owns the IP but is no IP protection advisor; (3) need for standardized, yet tailored processes; (4) spin-off's independence, yet further dependence on the university. The authors further determine implications from their study that help to overcome these tensions (Fink et al., 2023).

Selected Takeaways (the University's View)

- Technology transfer offices: Understand that a more effective strategy for technology commercialization is based on finding the entrepreneur who will be the vehicle of commercialization, not in promoting technology disclosures.
- Integrate technology transfer (entrepreneurship and IP management) as part of the "third mission" into the established structures and processes of the university.
- Solve the known tensions: E.g., use standardized, yet tailored processes.
- Start-ups: Reduce the time it takes to negotiate licenses by agreeing to standard and simple licenses that consider the unknown nature of start-ups.
- Don't spend all the time in the lab: Build the prototype and make sure the technology is sound and the IP is secured, but then get to market.

Part V: The Global IP Office's View

The Global IP Office's View addresses current developments and evolving institutional programs that support start-up IP activities. Authors provide insights from the Australian patent and trademark office IP Australia on the value of IP rights for start-ups, that has even a counsellor position for IP at their embassy in Beijing to guide and connect Australians to the China innovation system to help them make the most of their IP in China; the European Patent Office EPO with a Patent Academy that has developed a varied training and innovation support program to facilitate the commercialization of technology and high growth; as well as the French patent and trademark office INPI with an examination of the international start-up landscape and the United Nations lens on IP management.

Tools of the Trade: A Trading Nation's View on the Value of IP Rights for Start-Ups has been contributed by our authors Dr. Michael Falk, Office of the Chief Economist, IP Australia, Dr. Ben Mitra-Kahn, Program Manager, Australian Bureau of Statistics, Charlotte Trinh, Counsellor (Intellectual Property), Australian Embassy Beijing, and Dr. Haiyang Zhang, Economist, IP Australia. They summarize: For start-ups, effective use of intellectual property rights can help to attract resources, drive business value and growth, and take advantage of changes in market conditions. However, effective use of IP rights depends on start-ups seeking the right information and support, auditing their intellectual assets, understanding how the nature of their industry should inform their IP strategy, and drawing on the support of local partners in target markets. These benefits and approaches to IP are illustrated drawing on evidence from Australia about IP's role in entrepreneurship and business growth (Falk et al., 2023).

Market Success for Inventions: Patent Commercialization Scoreboard and Innovation Support by our authors Thomas Bereuter, Programme Area Manager Innovation Support at the European Patent Academy, Adéla Dvořáková, LESI High-growth Enterprise (HGE) Task Force, and Ilja Rudyk Senior Economist, all from the European Patent Office. They summarize: Deep tech start-up companies aiming for high growth must focus on intellectual property rights, particularly on patents. As demonstrated by the EPO's studies, patent protection makes it easier for these businesses to gain market access, strengthen their negotiation position, attract funding, or achieve strategic value in positioning the company. Start-ups and SMEs, along with universities and public research organizations, face similar challenges in commercializing their patented inventions, particularly the lack of resources, skills, and contacts. Identifying the right partners and their contacts, and the cost and complexity of negotiations, often limits collaborative exploitation. For this reason, the EPO's Patent Academy has developed a varied training program to facilitate the commercialization of technology and high growth. In particular, the EPO-LESI High-growth Technology Business initiative is tailored to enhancing the knowledge and skills of business decision-makers and IP professionals in deep tech start-ups and SMEs, thereby enabling them to maximize the market success of their patented inventions (Bereuter et al., 2023).

Raising IP Awareness of Start-Ups: A French Perspective has been contributed by Philippe Borne, Delegate of the French Patent and Trademark Office, INPI, France. He summarizes: Thanks to a high degree of active support from the public authorities, France became a leading start-up nation. This is especially the result of a realization by the country's policymakers that the ability to stay in the technology race was key to maintaining one's position in a tough world competition increasingly dominated by the rivalry between the United States and China. The early achievements manifested in the development of the French start-up ecosystem are not totally correlated with an intellectual property policy of the same magnitude. This puts the high-value French start-up community at potential risk, particularly vis-à-vis predator actions of global players in the field of disruptive technologies. An examination of the international landscape allows us to identify the possible reasons for such a situation as well as possible remedies (Borne, 2023).

Strategic Considerations and the Seed-to-Maturity Model when Establishing the Ideal IP Portfolio in the Start-Up Context has been contributed by Dr. T. Alexander Puutio, Intellectual Property Rights Advisor, from the United Nations Economic and Social Commission's Asia-Pacific Research Network on Trade (ARTNeT). He summarizes: His chapter integrates IP assets into the lean start-up method and explores the features and trade-offs of trade secrets, trademarks, copyrights, and patents in the start-up context. The utility and strategic value of these IP assets are analyzed against the seed-to-maturity framework and minimum-viable-product design process, and the chapter culminates in tangible strategies for maximizing the value of "lean IP portfolios" (Puutio, 2023).

Selected Takeaways (the Global IP Office's View)

- A very early incorporation of IP into your business strategy will greatly foster all your stages of development.
- An effective IP strategy begins by identifying your sources of intangible value, and how to support and build them using IP tools and drawing on expert advice.
- Technology start-ups should focus on IP, and patents in particular, from day one.
- Start-up founders should therefore aim to improve their IP knowledge and skills, and seek support.
- IP assets are relevant to all start-ups, at a minimum through trade secrets and trademarks.

Part VI: The Advisor's View

The Advisor's View provides an overview of the main types of intellectual property rights and serves as an introduction to the complementary use of formal and informal protection strategies to create and to capture the value of innovation. Authors advise what technology- or science-driven start-ups can do to secure and manage their intellectual property and how to avoid common pitfalls. Further insights are provided into the power of modern analytic tools and how these might be used by start-ups, and finally, a practical introduction is given on how to circumvent third-party patents in a structured way by applying the so-called TRIZ methodology.

An Introduction to Intellectual Property Rights and Formal and Informal Protection Strategies has been contributed by Dr. Martin A. Bader, Professor of Technology Management and Entrepreneurship at THI Business School from the Technical University of Applied Sciences Ingolstadt, Germany, and a European and Swiss patent attorney at the innovation and intellectual property advisory group BGW AG, St. Gallen, Switzerland. He provides an introduction to the most common types of intellectual property rights, such as patents, trademarks, designs, and more. He also highlights why only a holistic complementation of both de facto and legal protection strategies can provide effective innovation protection for competitive advantage and

to capture value. His chapter concludes with 10 practical tips for start-ups to manage protection strategies and IPRs (Bader, 2023).

Important Aspects of IP for Start-Ups: Strategies for Avoiding Common Pitfalls has been contributed by our authors Louis Lagler and Dr. Christian Ebner, both European and Swiss patent attorneys at Rentsch Partner Ltd., Zurich, Switzerland. They summarize: A technology- or science-driven start-up's intellectual property is one of its most crucial assets and thus securing its intellectual property is of utmost importance for the company's success. In their chapter, the authors draw from their own experience to highlight IP strategies and common pitfalls that start-ups face in securing their intellectual property (Lagler & Ebner, 2023).

Blind as a Mole? How Patent Data Can Help Start-Ups to Compete has been contributed by Dr. Sevim Süzeroglu-Melchiors, Professor of Digital Entrepreneurship and Finance at the Technical University of Applied Sciences Regensburg, Germany. She provides an overview about common types of patent search and analyses such as novelty search, freedom-to-operate search, IP landscapes, patent-based market and competitor analysis, technology scouting, and infringement assessment. A special section highlights the step-by-step approach for conducting an IP technology landscape. She concludes with practical tips for making the most out of patent data in a strategic and operative view (Süzeroglu-Melchiors, 2023).

Technical Patent Circumvention: How Start-Ups Can Get Around Infringed Patents has been contributed by our authors Prof. Dr. Oliver Mayer, Engineering Office Mayer, Munich, Germany, and Frank C. Schnittker, thl engineering, Damme, Germany. They summarize: When a company has a great idea to solve a technical problem, it usually gets disillusioned upon finding that this idea has already been patented by somebody else. Usually in this case one starts to activate lawyers in order to find out if one can legally get around the patent. A nullity complaint at court or negotiations on how to be able to use the patent through license fees or cross-licensing are options. There is a technical way to circumvent patents as well. The authors describe a systematic approach and a procedure on how R & D departments can work with existing patents and circumvent them with technology. This procedure is not only valid for circumventing competitors' patents but also for fostering one's own patent applications and thus developing one's own solid patenting strategy (Mayer & Schnittker, 2023).

Selected Takeaways (the Advisor's View)

- IP is not an end in itself: Amalgamate both, business and IP strategies, to gain and sustain competitive advantage.
- For IPR legal procedures, such as filings, registrations, opposition, nullity, or enforcement proceedings, it is advisable to engage an IP attorney.
- Find the right balance between your budget and scope of protection.
- Prevention is better than healing.
- IP data is a useful source for start-ups along all four venture phases.

Table 1 Overview of covered topics, industries, technologies, regions, and countries by the six stakeholder views (authors' own compilation)

Point of view (stakeholder)	Topics covered	Industries/technologies	Regions/countries[a]
I. Start-up	• Learnings from the Israel start-up ecosystem • Learnings from German sustainable technologies start-ups • Learnings from the Australian start-up ecosystem • Learnings from the start-up ecosystem 'Silicon Island' in Brazil	• Cybersecurity • IT and software • Mining • Climate tech • Sustainable technologies	• Australia • Brazil • Germany • Israel
II. Investor	• Insights of business angels, VCs, and investors • Role of IP and intangible assets in tech private placements • Crossroads between investment and IP • IP-backed financing • Monetary IP valuation methods • Definitions of intellectual assets (IA) and intellectual property (IP) • The 'ladder of IP savviness' • The 'minimum viable IP (MVIP)' management strategy • The dos and don'ts of IP protection and investor attractiveness	• Artificial intelligence (AI) • Machine learning (ML) • Software • Biotech • Consumer products • Food/nutrition • Pharma • Digital/health • Medtech • Medical instruments • Orthopedics • Sports/wellness	• Canada • Israel • Silicon Valley • Switzerland • United States
III. Corporation	• IP as an important value driver for start-up strategies in established companies (Siemens) • Applying IP learnings from large companies to start-ups (Nokia) • How the life sciences industry handles IP in start-ups (Bayer) • The Japanese start-up ecosystem and corporate culture • Strategic uses of IP and IP strategy implementations in start-ups • IP value creation, portfolio construction and strategy • IP risk identification and mitigation strategy	• Automation and industrialization • Internet of things (IoT) • Industry 4.0 • Information and communications technologies • Automotive • Consumer electronics • Life sciences • Pharma • Digital/health • Crop science/agrobusiness • Social media	• Finland • Germany • Japan • Switzerland • United States • International

IV. University	• Academic entrepreneurship • Technology transfer and commercialization • Royalty rates • Cooperative research and development agreements • The 'triple helix': university-industry-government interaction • Incubator programs • Tensions and implications for university spin-offs and TTOs	• Computer science • Software as a service (SaaS) • Biotech • Pharma • Oncoceutics	• Germany • Israel • Singapore • South Africa • Switzerland • Thailand • United States
V. Global IP Office	• Raising IP awareness of start-ups • IP as a tool of the trade and export • Policy makers and public authorities • The 'four level IP strategy model' • Patent commercialization scoreboard and innovation support • Strategic considerations when establishing the ideal IP portfolio in the start-up context	• Artificial intelligence (AI) • Blockchain • Quantum computing • Cybersecurity • Deep tech • Medtech • Medical instruments • Biotech • Cleantech • Autonomous driving • Mechanical engineering	• Australia • Brazil • China • France • India • Japan • Korea • United States • Europe • International
VI. Advisor	• Introduction to the different types of IPRs • Formal and informal protection strategies • How to capture the value of innovation with IP (the 'St. Gallen patent management model') • How to secure IP in technology- or science-driven start-ups • How to use patent searches and analyses • How to bypass a patent in a structured way: the 'TRIZ methodology'	• Technology • Science • Quantum computing • Biodegradable packaging • Medtech • Lighting • Wind energy	• The 'Big Five' world IP regions • Germany • Switzerland • Europe • International

[a]In alphabetical order

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